Investigating Molecular Transport in the Human Brain from MRI

Physics-informed neural networks for estimating tracer diffusivity
Zapf et al. - *Scientific Reports* (2022), 12:15475

Core question. Can a physics-informed neural network (PINN) recover an apparent diffusion coefficient governing long-term molecular transport in human brain white matter from sparse, noisy MRI observations - and can it agree with a conventional finite-element inverse-problem solution?

1. Scientific and clinical motivation

The study is motivated by the brain's glymphatic/waste-clearance system. The authors analyze the spread of intrathecally administered gadobutrol tracer over roughly two days using T1-weighted contrast-enhanced MRI. Understanding transport and clearance is relevant because impaired waste clearance has been linked to neurological disease, while the precise mechanisms of glymphatic transport remain debated.

The paper combines **medical imaging**, **partial differential equations**, and **machine learning**. The central inverse problem is to infer a physiological parameter from a small number of spatially complex MRI snapshots.

2. Mathematical problem

Tracer concentration c(x,t) is modeled with the diffusion equation **dc/dt = D Laplacian(c)**, where D is an unknown positive apparent diffusion coefficient. Concentration measurements are available at only four times: 0, 7, 24 and 46 hours. The analysis is restricted to a three-dimensional white-matter subregion.

The model assumes a spatially constant scalar D. The authors acknowledge that brain diffusion can be anisotropic, but their patient-specific analysis supports the approximation as reasonable for the selected region.

3. End-to-end MRI-to-parameter workflow

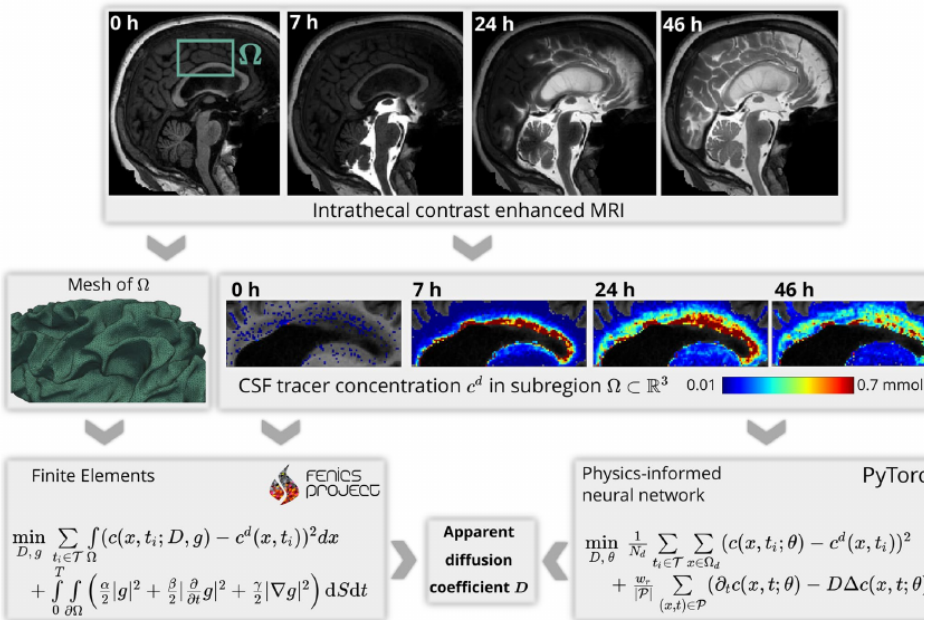


Figure 1. Flowchart illustrating our workflow from clinical images to estimated tracer diffusivity in the hu brain. From the FreeSurfer[45] segmentation of a baseline MRI at $t = 0$, we define and mesh a subregion Ω of the white matter. Intrathecal contrast enhanced MRI at later times $t = 7,\ 24,\ 46$ h are used to estimate the concentration of the tracer in the subregion. We then use both a finite element based approach and physics-informed neural networks to determine the scalar diffusion coefficient that describes best the concentration dynamics in Ω.

Article Figure 1 (p. 3). Baseline and post-contrast MRI are converted into tracer concentration data in a white-matter subregion. Finite elements and a PINN then estimate the same scalar diffusion coefficient D.

The baseline MRI is segmented with FreeSurfer to define the white-matter region. Subsequent contrast-enhanced images provide tracer concentrations. FEM requires a mesh and an explicitly solved PDE-constrained optimization problem. The PINN represents c(x,t) by a fully connected neural network and incorporates the diffusion equation directly into its training objective.

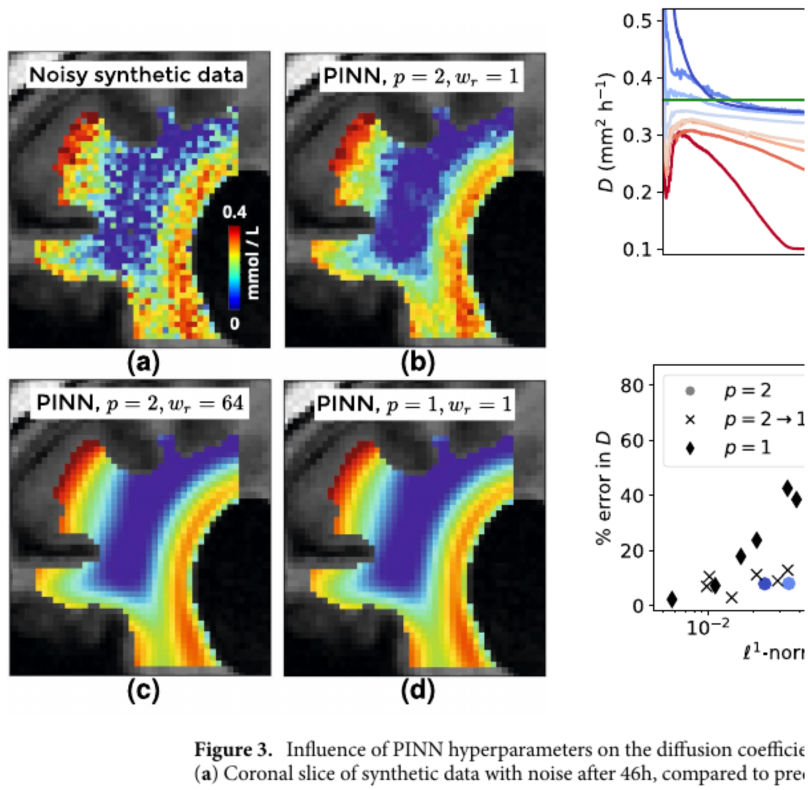
4. PINN formulation versus FEM

Feature	Physics-informed neural network	Finite-element approach
State representation	Neural network c(x,t;theta)	Numerical PDE solution on a mesh
Physics enforcement	PDE residual penalty in loss	PDE explicitly solved in weak/discrete form
Data term	Mismatch to MRI concentration samples	Mismatch to MRI concentration samples
Boundary handling	Boundary condition omitted from PINN loss	Boundary value treated as a control variable
Unknown parameter	D trained jointly with network weights	D optimized in PDE-constrained problem
Main practical issue	Non-convex training and hyperparameter sensitivity	Meshing, regularization and repeated PDE solves

The PINN objective combines a data-discrepancy loss with a weighted PDE-residual loss. The PDE weight, residual norm (L1 or L2), learning rate, parameterization of D and adaptive sampling strategy all affect whether training recovers the correct physical parameter.

5. Synthetic-data validation and the problem of noise

With noise-free synthetic data, both PINN and FEM recover the known diffusion coefficient within a few percent. Noisy synthetic data reveal a major failure mode: a standard PINN can **overfit measurement noise**, satisfy the data too closely, and drive D toward its imposed lower bound rather than the true value.

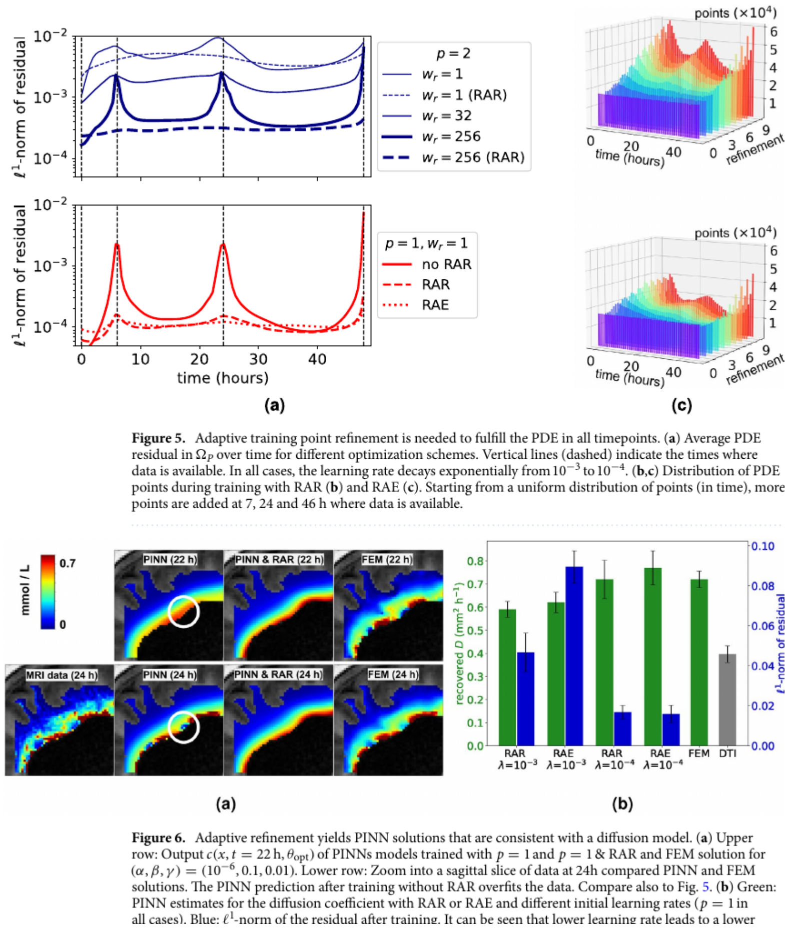


Article Figure 3 (p. 4). A weak physics penalty permits overfitting. Increasing the PDE-loss weight or using an L1 residual suppresses the noisy fit. The lower-right plot shows that smaller PDE residuals coincide with more accurate recovery of D.

For the L2 formulation, increasing the PDE weight to roughly 64 or above brings the recovered coefficient to within about 10% of the known value. The L1 norm is another effective remedy because it is less sensitive to outliers. Across the tested settings, the post-training PDE residual is a useful indicator of parameter quality.

6. Clinical MRI results and adaptive refinement

Real MRI is harder than the synthetic case because the diffusion model is only an approximation and measured tracer concentration varies along the domain boundary. The PINN was trained for 100,000 epochs with ADAM and learning-rate decay.



Article Figures 5-6 (p. 7). Residual-based adaptive refinement (RAR) and residual-based adaptive exchange (RAE) concentrate PDE training points in difficult space-time regions, reduce residual spikes near observation times, and prevent the PINN from simply overfitting the measured images.

7. Quantitative findings

Comparison	Estimated D (mm2 h-1)	Interpretation
Main PINN setting (L1)	0.75	Close to the FEM reference
FEM reference	0.72	Benchmark from the finite-element analysis
Patient NPH2 - PINN / FEM	0.48 / 0.50	Very close agreement
Patient REF - PINN / FEM	0.41 / 0.50	Larger discrepancy, same order of magnitude
Overall MRI conclusion	~0.6-0.7	PINN and FEM give similar ADC estimates when properly configured
DTI comparison	~0.4	Lower than the long-timescale MRI/FEM/PINN estimate

With a suitable setup, the authors report that PINNs are approximately as accurate as FEM and their GPU implementation is roughly twice as fast as the FEM implementation used for the MRI data. The speed comparison is implementation-specific, but it supports the potential of PINNs for semi-automated analysis across patient cohorts.

8. Why adaptive residual sampling matters

RAR adds PDE collocation points where the residual is high, improving physics consistency but increasing computation by about 25% in the reported setting. RAE instead adds high-residual points while removing points where the PDE is already well satisfied, keeping the point count and computing time approximately constant. Both produced similarly low residuals.

9. Limitations

The inverse problem is nonlinear and ill-posed, with sparse and noisy measurements. PINN training can converge to poor local minima and is sensitive to loss weights, norms, learning rates and parameterization. The scalar homogeneous diffusion model is physiologically simplified, and the limited patient data prevent broad uncertainty analysis.

10. Overall conclusion

The key result is that reliable parameter recovery depends on forcing the trained neural network to satisfy the governing PDE with a sufficiently small residual. When that happens, the PINN estimate approaches the established finite-element result. L1 PDE penalization and adaptive residual sampling are central techniques for achieving this with noisy clinical MRI.

The study is a strong example of **scientific machine learning**: MRI provides patient-specific observations, a PDE provides mechanistic structure, and deep learning supplies a flexible mesh-free surrogate for the inverse problem.

Source: Zapf B, Haubner J, Kuchta M, Ringstad G, Eide PK, Mardal K-A. "Investigating molecular transport in the human brain from MRI with physics-informed neural networks." *Scientific Reports*. 2022;12:15475. Figures reproduced as crops from the supplied article for study and explanatory purposes.